MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, LOW-POWER SCHOTTKY TTL,
ARITHMETIC LOGIC UNITS/FUNCTION GENERATORS,
MONOLITHIC SILICON

This amendment forms a part of MIL-M-38510/308A, dated 2 February 1982, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* Title: Delete and substitute title as printed on this amendment.

1.1, line two: Delete "Three" and substitute "Two".

* 1.2.3, case outline delete the following:

"L F-7 (24-lead, 3/8" x 1/2", flat package)".

Add the following:

"3 C-4 (28-terminal, .450" x .450"), square chip carrier package".

* 1.3, thermal resistance: Delete "0.04°C/mW for flat package;
0.02°C/mW for dual-in-line package"

and substitute "Case 3 - - - - - - - - - - - - - 0.08°C/mW 2/
Cases J, K, Z - - - - - - - - - (See MIL-M-38510, appendix C)"

* 1.3, Junction temperature, after (T_J) add "3/".

* Add at bottom of page:

"2/ When a thermal resistance value is included in MIL-M-38510, appendix C, it shall supersede the values stated herein.

"3/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions, as per method 5004 of MIL-STD-883."

PAGE 2

* 3.2.5, Schematic circuits, delete and substitute:

"3.2.5 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in the specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained by the agent activity and will be available upon request."

PAGE 3

TABLE I, conditions column heading: Add "$-55°C \leq T_C \leq +125°C$". Make this same change on pages 4 through 6.

PAGE 6

TABLE I, conditions column heading: Add "1/".

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PAGE 7

* 3.5, delete and substitute:

"3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III."

* 3.6, delete the second sentence.

* TABLE II, test requirement: Delete "Interim" and substitute "Initial".

TABLE II, group B test requirements: Add "Subgroup 5".

TABLE II: Delete class C device column in its entirety.

* 4.2b: Delete "Interim" and substitute "Initial" (2 places).

PAGE 8

* 4.2c, delete and substitute:

"c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510."

* 4.4.3, delete and substitute:

"4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical tests shall be as specified in table II herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions, or equivalent:

(1) Test condition D or E, using the circuit shown on figure 5, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by method 1005 of MIL-STD-883."

* 4.6, delete in its entirety.

PAGE 9

* FIGURE 1: Delete "Cases J, K, L, and Z" and substitute "Cases J, K, Z". Add new figure for case 3 as printed on page 3 of this amendment.

Pin 1: Delete "B_0" and substitute "$\overline{B}_0$".
Pin 2: Delete "A_0" and substitute "$\overline{A}_0$".
Pin 9: Delete "F_0" and substitute "$\overline{F}_0$".
Pin 10: Delete "F_1" and substitute "$\overline{F}_1$".
Pin 11: Delete "F_2" and substitute "$\overline{F}_2$".
Pin 17: Delete "$\overline{G}$" and substitute "G".

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Device type 01

Case 3

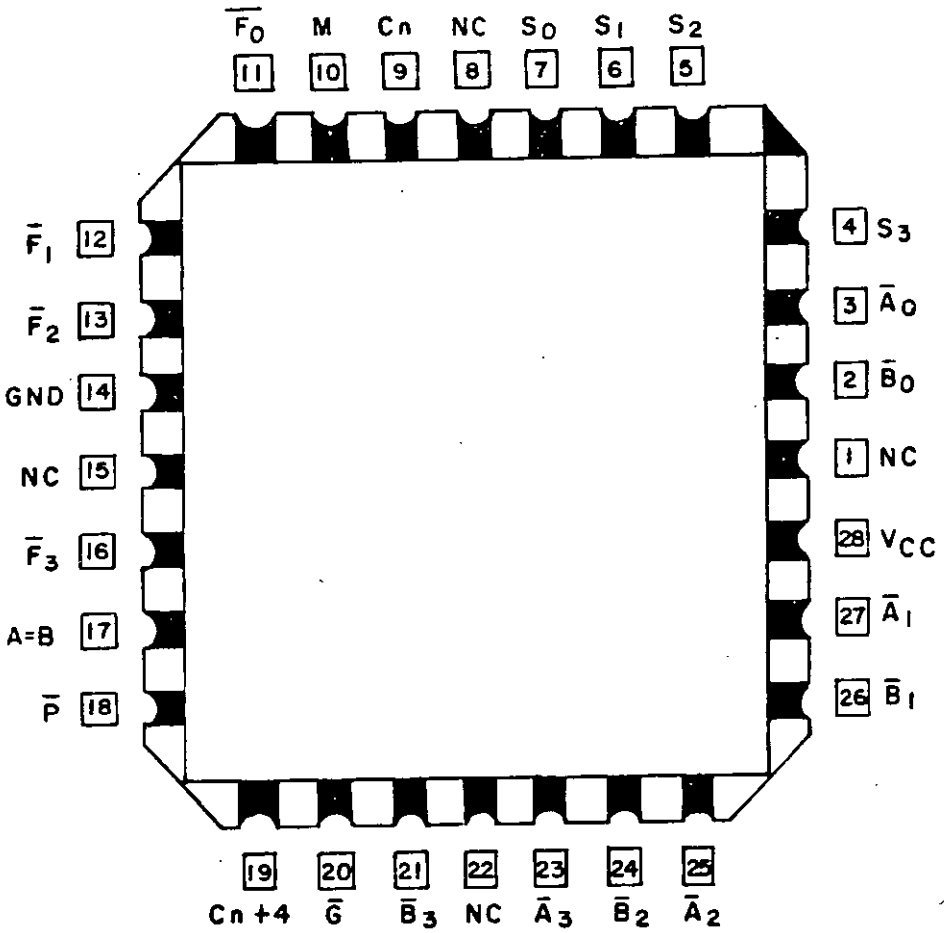


FIGURE 1. Terminal connections (top view).

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PAGE 15

* TABLE III, device type 01, under cases, delete pin designations and substitute the following: Make these same changes on pages 16 through 20.

Cases	Pins not referenced are NC																							
J,K,Z	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
3	2	3	4	5	6	7	9	10	11	12	13	14	16	17	18	19	20	21	23	24	25	26	27	28
Test no.	$\overline{B}_0$	$\overline{A}_0$	S_3	S_2	S_1	S_0	C_n	M	F_0	F_1	F_2	GND	F_3	A=B	P	C_{n+4}	G	$\overline{B}_3$	$\overline{A}_3$	$\overline{B}_2$	$\overline{A}_2$	$\overline{B}_1$	$\overline{A}_1$	V_{CC}

PAGE 20

Footnote 4/: Delete "115" and substitute "120".

PAGE 21

* 6.1, Notes: Delete in its entirety.

* 6.2, delete and substitute:

"6.2 Intended use. Microcircuits conforming to this specification are intended for use for Government microcircuit applications (original equipment) and logistic purposes."

* 6.4, delete "GND - - - - - Electrical ground (common terminal)". and substitute "GND - - - - - Ground zero voltage potential".

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-0371)

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